

Panel Data

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Summary

Gershenson, S., Holt, S. B., & Papageorge, N. W. (2016). Who believes in me? The effect of student–teacher demographic match on teacher expectations. *Economics of Education Review*, 52, 209–224.

- **Question:** student–teacher demographic mismatch ? $\Rightarrow$ teachers' expectations about a student's ultimate educational attainment?
- **Identification:** each student is observed *twice*, so the paper turns the problem into a short panel fixed-effects design.
- **Main finding:** relative to black teachers, non-black teachers have significantly lower expectations for black students; effects are larger for black male students and in math.

Data (Section 3)

- Unit of observation in estimation: **student–subject dyad**, (i, s) .
- For each student i , there are exactly two observations:

$$s \in \{M, R\} \quad (\text{math teacher, reading teacher}).$$

- Outcomes are *binary indicators* built from teachers' reported expectations:
 - $Y_{is}^H = 1\{\text{teacher expects} \leq \text{high school diploma}\}$,
 - $Y_{is}^C = 1\{\text{teacher expects} \geq \text{four-year college degree}\}$.
- Key regressors:
 - **Other**_{is}: student–teacher **other-race** indicator, student–teacher **other-sex** indicator,
 - **x**_{is}: teacher controls: experience, experience squared, graduate degree, major in subject taught.
- Note: this is like a panel with $T = 2$, where the within-unit dimension is *subject*, not calendar time.

Problems with pooled regression

- A naive model would compare *different* students:

$$Y_{is} = \gamma_0 + \gamma_1 \text{OtherRace}_{is} + \gamma_2 \text{OtherSex}_{is} + u_{is}.$$

- But

$$\mathbb{E}[u_{is} \mid \text{OtherRace}_{is}, \text{OtherSex}_{is}] \neq 0$$

- Why? Students assigned to demographically different teachers differ systematically in:
 - prior achievement,
 - family socioeconomic status and parental education,
 - aspirations and behavior,
 - school context and course placement.
 - ...

Baseline identification strategy (Section 4)

$$E_{is} = \alpha_s + \theta_i + \mathbf{x}_{is}'\beta + \text{Other}_{is}'\delta + \varepsilon_{is}, \quad s \in \{M, R\}.$$

- E_{is} : teacher s 's expectation for student i .
- α_s : **subject fixed effect** (math vs. reading).
- θ_i : **student fixed effect**.
- $\mathbf{x}_{is}$: observed teacher characteristics.
- Other_{is} : mismatch variables:
 - baseline: other-race, other-sex;
 - extension: four mutually exclusive cells (same race/same sex omitted).
- δ : parameter of interest.

Why include subject fixed effects?

- **Intuition:** math and reading teachers may systematically differ in how they form expectations, even for the same student.
 - Example 1: math teachers may be more conservative in forecasting college completion.
 - Example 2: reading teachers may observe behaviors or skills more tightly linked to college-going.
- If we omit α_s , then mismatch can pick up *subject-specific average differences* rather than bias from demographic mismatch.
- With only two subjects, α_s is doing an important but limited job:
 - it absorbs the *average* math-versus-reading difference,
 - **it does *not* absorb student-specific comparative advantage in one subject.**
- That second issue becomes the remaining identification threat later.

Why include individual (student) fixed effects?

- **Intuition:** some students are simply more likely to elicit high expectations from *all* teachers.
- This includes both observables and unobservables:
 - ability, motivation, behavior,
 - family support,
 - perceived barriers to college,
 - soft information known to both teachers.
- If we compare student A with a white teacher to student B with a black teacher, any expectation gap may reflect $A \neq B$, not teacher mismatch.
- θ_i absorbs all student-specific time-invariant factors shared across the two teacher reports.

The remaining threat: subject-specific endogenous sorting

- Student FE does *not* solve everything.
- The remaining threat is sorting that varies **by subject and by student background**.

$$\mathbb{E}[\Delta\varepsilon_i \mid \Delta\text{Other}_i, \Delta x_i] = 0 \quad \text{may fail.}$$

- **Intuitive example:** suppose low-math-ability black students are systematically assigned to white math teachers, while there is no comparable sorting pattern in reading.
- Then, within the same student, the white math teacher may look “more pessimistic” partly because of subject-specific sorting, not because of difference in race per se.
- (Panel language) this is like **time-varying omitted variables** that survive the unit FE transformation.

Additional Questions

$$E_{is} = \alpha_s + \theta_i + x'_{is}\beta + \text{Other}'_{is}\delta + \varepsilon_{is}, \quad s \in \{M, R\}.$$

- ① Why not control for observable student characteristics?
- ② Why not include school fixed effect?
- ③ Why not include teacher fixed effect?

Growth Empirics: A Panel Data Approach

- **Question:** How should we estimate cross-country growth convergence when countries differ in persistent, unobserved ways?
- **Background** Mankiw, Romer, and Weil (1992) estimate a Solow-based *single cross-section* regression.
- **Islam (1995)** reinterpret the same convergence equation as a **dynamic panel model** with country fixed effects.

pooled/cross-section OLS $\Rightarrow$ omitted country heterogeneity

panel FE $\Rightarrow$ control for differences in the aggregate production function across economies

- **Results**
 - estimated conditional convergence is much faster;
 - estimated elasticity of output to capital becomes more plausible;

Solow model in effective labor units

Start from the Cobb–Douglas production function with labor-augmenting technical progress:

$$Y(t) = K(t)^\alpha (A(t)L(t))^{1-\alpha}, \quad 0 < \alpha < 1,$$

with exogenous growth

$$L(t) = L(0)e^{nt}, \quad A(t) = A(0)e^{gt}.$$

Define variables per unit of effective labor:

$$k(t) \equiv \frac{K(t)}{A(t)L(t)}, \quad \tilde{y}(t) \equiv \frac{Y(t)}{A(t)L(t)}.$$

Then

$$\tilde{y}(t) = k(t)^\alpha.$$

With constant saving rate s and depreciation δ , capital accumulation becomes

$$\dot{k}(t) = s\tilde{y}(t) - (n + g + \delta)k(t) = sk(t)^\alpha - (n + g + \delta)k(t).$$

Steady state solves $\dot{k} = 0$:

$$k^* = \left(\frac{s}{n + g + \delta} \right)^{\frac{1}{1-\alpha}}, \quad \tilde{y}^* = (k^*)^\alpha = \left(\frac{s}{n + g + \delta} \right)^{\frac{\alpha}{1-\alpha}}.$$

M-R-W steady-state level regression

Per *worker* income is

$$y(t) \equiv \frac{Y(t)}{L(t)} = A(t)\tilde{y}(t) = A(0)e^{gt}\tilde{y}(t).$$

Hence at the steady state,

$$y^*(t) = A(0)e^{gt} \left(\frac{s}{n+g+\delta} \right)^{\frac{\alpha}{1-\alpha}}.$$

Taking logs:

$$\ln y^*(t) = \ln A(0) + gt + \frac{\alpha}{1-\alpha} \ln s - \frac{\alpha}{1-\alpha} \ln(n+g+\delta).$$

For a single cross section at a common date t , gt is absorbed into the constant, so

$$\ln y_i = a + \frac{\alpha}{1-\alpha} \ln s_i - \frac{\alpha}{1-\alpha} \ln(n_i + g + \delta) + \varepsilon_i,$$

where

$$\varepsilon_i \equiv \ln A_i(0) - \mathbb{E}[\ln A_i(0)]. \text{ Assume: } \varepsilon_i \perp \{\ln s_i, \ln(n_i + g + \delta)\}.$$

Interpretation:

- s_i and n_i proxy for cross-country differences in steady states;
- ε_i bundles technology, institutions, resources, climate, etc (persistent country-specific heterogeneity)

Transition dynamics

Near the steady state, linearization of the Solow dynamics implies

$$\frac{d \ln \tilde{y}(t)}{dt} = \lambda (\ln \tilde{y}^* - \ln \tilde{y}(t)), \quad \lambda = (n + g + \delta)(1 - \alpha).$$

Solving over an interval of length $T = t_2 - t_1$:

$$\ln \tilde{y}(t_2) = (1 - e^{-\lambda T}) \ln \tilde{y}^* + e^{-\lambda T} \ln \tilde{y}(t_1).$$

Equivalently,

$$\ln \tilde{y}(t_2) - \ln \tilde{y}(t_1) = (1 - e^{-\lambda T}) (\ln \tilde{y}^* - \ln \tilde{y}(t_1)).$$

Substitute

$$\ln \tilde{y}^* = \frac{\alpha}{1 - \alpha} \ln s - \frac{\alpha}{1 - \alpha} \ln(n + g + \delta),$$

to obtain

$$\ln \tilde{y}(t_2) - \ln \tilde{y}(t_1) = (1 - e^{-\lambda T}) \frac{\alpha}{1 - \alpha} \ln s - (1 - e^{-\lambda T}) \frac{\alpha}{1 - \alpha} \ln(n + g + \delta) - (1 - e^{-\lambda T}) \ln \tilde{y}(t_1).$$

Key point: the initial-income coefficient identifies convergence:

$$\text{coefficient on initial income} = -(1 - e^{-\lambda T}) < 0.$$

Towards dynamic panels

Since

$$\tilde{y}(t) = \frac{Y(t)}{A(t)L(t)} = \frac{y(t)}{A(0)e^{gt}},$$

we have

$$\ln \tilde{y}(t) = \ln y(t) - \ln A(0) - gt.$$

Substitute this into the previous equation. After rearranging,

$$\begin{aligned} \ln y(t_2) = & \underbrace{e^{-\lambda T}}_{\gamma} \ln y(t_1) + \underbrace{(1 - e^{-\lambda T}) \frac{\alpha}{1 - \alpha}}_{\beta_1} \ln s - \underbrace{(1 - e^{-\lambda T}) \frac{\alpha}{1 - \alpha}}_{\beta_2} \ln(n + g + \delta) \\ & + \underbrace{(1 - e^{-\lambda T}) \ln A(0)}_{\mu_i} + g(t_2 - e^{-\lambda T} t_1). \end{aligned}$$

Let $z_{it} \equiv \ln y_i(t)$. Then the convergence equation becomes

$$z_{it} = \gamma z_{i,t-1} + \beta_1 \ln s_{it} + \beta_2 \ln(n_{it} + g + \delta) + \eta_t + \mu_i + v_{it},$$

Note: can recover structural parameters (λ, α) under restrictions.

More details

Why not random effects?

- Random effects requires μ_i uncorrelated with regressors.
- But *that correlation is exactly the problem Islam (1995) wants to address.*

Why not cross-section IV?

- In principle: instrument $\ln s_i$ and $\ln(n_i + g + \delta)$.
- In practice: hard to find instruments correlated with those variables but orthogonal to broad latent $A_i(0)$.

Why FE estimator?

- Nickell bias.
- Islam's defense: for this application, T -asymptotics and Monte Carlo evidence suggest it performs OK.

Data

- samples close to M-R-W: NONOIL (96), INTER (74), OECD (22)
- five-year intervals over 1960–1985;
- $g + \delta = 0.05$;
- n is total population growth (not working-age population growth).

Results

Sample	Single cross section			FE Estimator		
	NONOIL	INTER	OECD	NONOIL	INTER	OECD
$\hat{\lambda}$	0.0056	0.0092	0.0153	0.0467	0.0458	0.0926
$\hat{\alpha}$	0.8346	0.7628	0.6036	0.4398	0.4575	0.2047

Discussion: omitted-variable-bias

$$z_{it} = \gamma z_{i,t-1} + \beta x_{it} + \eta_t + \mu_i + v_{it}.$$

Sign logic for γ : μ_i enters the full model positively; countries with high $A_i(0)$ also tend to have high initial income $z_{i,t-1}$; therefore omitting μ_i tends to bias $\hat{\gamma}$ upward.

Since

$$\lambda = -\frac{1}{T} \ln \gamma,$$

so an upward bias in $\hat{\gamma}$ means a **downward bias** in estimated convergence.

Sign logic for α :

$$\alpha = \frac{\beta}{1 - \gamma + \beta}.$$

Holding β fixed, a larger γ raises α . Thus the same omitted variable problem also helps explain why cross-section and pooled regressions overstate the capital elasticity.

Exercise I

Suppose individuals are permanently nested in states: individual i belongs to state $s(i)$. The panel is balanced, so each individual is observed in every period $t = 1, \dots, T$. Suppose the treatment varies only at the state-time level:

$$D_{it} = D_{s(i)t}.$$

Show that the FE estimate of θ is identical in the two regressions

$$Y_{it} = \alpha_{s(i)} + \delta_t + \theta D_{s(i)t} + \varepsilon_{it}, \quad (1)$$

and

$$Y_{it} = u_i + \delta_t + \theta D_{s(i)t} + \varepsilon_{it}. \quad (2)$$

Hint: apply the Frisch–Waugh–Lovell theorem.

Card and Krueger, 1994

- **Research question:** How minimum wage increase affects employment
- **Policy Change:** On April 1, 1992 minimum wage in NJ rose from \$4.25 to \$5.05
- **Research Design:** Diff-in-Diff, comparing employment growth in fast-food restaurants in NJ (treatment group) and PA (control group)
- **Data:** Survey of 410 restaurants in NJ and PA before and after policy change
- **Results:** No evidence that minimum wage increase reduces employment

Card and Krueger (1994): Setup and Notation I

- Stores are indexed by i .
- State indicator:

$$G_i = \mathbf{1}\{S_i = \text{NJ}\}, \quad G_i = 0 \text{ for PA.}$$

- Time periods:

$$t = 1 \text{ (before, February 1992)}, \quad t = 2 \text{ (after, August 1992).}$$

- Observed employment:

$$Y_{it}.$$

- Store covariates:

$$X_i.$$

Card and Krueger (1994): Setup and Notation II

- Potential outcomes:

$Y_{it}(1)$ = employment if the higher minimum wage applies by period t ,

$Y_{it}(0)$ = employment if the minimum wage is not increased by period t .

- Because the policy applies only in NJ and only after April 1992,

$$Y_{i1} = Y_{i1}(0) \text{ for all } i,$$

$$Y_{i2} = \begin{cases} Y_{i2}(1), & G_i = 1, \\ Y_{i2}(0), & G_i = 0. \end{cases}$$

- Target parameter (ATT):

$$\theta_0 = \mathbb{E}[Y_{i2}(1) - Y_{i2}(0) \mid G_i = 1].$$

Question 1: Basic DiD Estimand

Define the population DiD estimand

$$\theta_{\text{DiD}} = \left(\mathbb{E}[Y_{i2} \mid G_i = 1] - \mathbb{E}[Y_{i1} \mid G_i = 1] \right) - \left(\mathbb{E}[Y_{i2} \mid G_i = 0] - \mathbb{E}[Y_{i1} \mid G_i = 0] \right).$$

- 1 Write the sample analog of θ_{DiD} using cell means.
- 2 Using the sample averages

	$t = 1$ (before)	$t = 2$ (after)
PA	6	5
NJ	10	7

compute the numerical DiD estimate.

Question 2: Identification of the ATT

Under the notation above, give assumptions that are necessary and sufficient for

$$\theta_{\text{DiD}} = \theta_0.$$

Then:

- 1 interpret the assumption(s) in words;
- 2 explain why the key identifying condition is generally not directly testable from the observed data.

Question 3: Regression Implementations of DiD

Let $\Delta Y_i = Y_{i2} - Y_{i1}$ and $\text{Post}_t = \mathbf{1}\{t = 2\}$. Consider

$$\Delta Y_i = \alpha_0 + \theta_{\text{FD}} G_i + u_i,$$

and

$$Y_{it} = \beta_0 + \beta_1 G_i + \beta_2 \text{Post}_t + \theta_{\text{TW}} (G_i \cdot \text{Post}_t) + e_{it}.$$

- 1 Write the population OLS estimands for θ_{FD} and θ_{TW} .
- 2 Show that, under the appropriate mean-independence conditions on u_i and e_{it} ,

$$\theta_{\text{FD}} = \theta_{\text{TW}} = \theta_{\text{DiD}}.$$

Question 4: What Trends Are Identified?

Using the regression in Question 3, determine whether the following objects are identified:

- ① the average untreated trend

$$\mathbb{E}[Y_{i2}(0) - Y_{i1}(0)];$$

- ② the average policy trend for treated stores

$$\mathbb{E}[Y_{i2}(1) - Y_{i1}(1) \mid G_i = 1].$$

When an object is identified, express it in terms of $\beta_0, \beta_1, \beta_2, \theta_{TW}$. If an additional assumption is needed, state it clearly.

Question 5: Covariate-Adjusted DiD

Now consider the covariate-adjusted first-difference regression

$$\Delta Y_i = \gamma_0 + X_i' \gamma + \theta_X G_i + v_i, \quad \mathbb{E}[v_i \mid G_i, X_i] = 0.$$

- 1 Give economically meaningful conditions under which

$$\theta_X = \theta_0.$$

- 2 Discuss whether these conditions are weaker than the conditions in Question 2.

Question 6: Store Closures and Attrition

Let $R_i = 1$ if store i is still open in period 2, and $R_i = 0$ otherwise. Suppose one estimates

$$\theta_{CC} = \left(\mathbb{E}[Y_{i2} \mid G_i = 1, R_i = 1] - \mathbb{E}[Y_{i1} \mid G_i = 1, R_i = 1] \right) - \left(\mathbb{E}[Y_{i2} \mid G_i = 0, R_i = 1] - \mathbb{E}[Y_{i1} \mid G_i = 0, R_i = 1] \right).$$

- 1 State sufficient assumptions for $\theta_{CC} = \theta_0$.
- 2 Evaluate whether these assumptions are plausible.
- 3 If you find them too strong, propose an alternative strategy for learning about θ_0 when some stores close before period 2.

Question 7: NJ-Only Before–After Design

Using only NJ stores, consider the before–after estimator

$$\theta_{BA} = \mathbb{E}[Y_{i2} \mid G_i = 1] - \mathbb{E}[Y_{i1} \mid G_i = 1].$$

- 1 Under what assumptions is θ_{BA} a consistent estimator of θ_0 ?
- 2 Compare these assumptions with the assumptions required for the DiD design.

Question 8: Alternative IV Design I

Suppose, hypothetically, Card and Krueger only sampled PA stores. Within PA, some stores are randomly offered an incentive to voluntarily adopt the higher minimum wage.

Let

$$Z_i = \mathbf{1}\{\text{store } i \text{ is offered the incentive}\},$$

$$D_i = \mathbf{1}\{\text{store } i \text{ actually adopts the higher minimum wage}\},$$

and let Y_i denote post-policy employment.

- 1 Define an IV estimator for the effect of the higher minimum wage using Z_i as an instrument for D_i .
- 2 State assumptions under which this estimator identifies a LATE.
- 3 Explain why this IV estimand can differ from θ_0 and from the DiD estimand.